

Machine Learning Problem Definition Document

HealthConnect Project Kickoff and Problem Understanding

Experience Lab Internship Programme

Prepared by: Ayden Demanou, Junior Data Scientist, AnalystLab Africa Consulting

1. Introduction and Track Role

HealthConnect Clinic is a fictional healthcare provider facing recurring missed appointments (“no-shows”), inefficient use of appointment slots, and repetitive patient enquiries. The central project question is:

“How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?”

As the Data Science intern on this project, my responsibility in Week 4 is to translate this business problem into a well-scoped machine learning problem: **to assess whether the available appointment data can plausibly support a no-show prediction solution, and to define the target variable, candidate features, and an initial modelling approach that later stages (development, testing, refinement) will build on.**

2. Business Problem → Machine Learning Problem

2.1 Business Problem

- Patients frequently miss scheduled appointments, with no systematic way of knowing in advance which appointments are at risk.
- Missed appointments result in wasted clinical capacity, longer waiting lists, and lost administrative time.
- The clinic currently reacts to no-shows after they happen rather than intervening proactively (e.g., through targeted reminders or overbooking strategies).

2.2 Machine Learning Framing

The business problem can be reframed as a **supervised binary classification problem**: given information known at (or shortly after) the time an appointment is booked, predict whether a patient is likely to attend or miss that appointment. A reliable prediction would allow HealthConnect Clinic to:

- Prioritise reminder calls, SMS, or follow-up for high-risk appointments.
- Inform selective overbooking or waitlist strategies for slots at high no-show risk.

This is consistent with well-established literature on healthcare appointment no-show prediction, where similar demographic, scheduling, and historical-behaviour variables are typically the strongest predictors.

3. Initial Data Assessment

3.1 Dataset Overview

The HealthConnect_Appointment_Data.csv file (5,000 appointment records, 18 columns) has now been reviewed and profiled. Full exploratory code, outputs, and charts are provided in the accompanying notebook, HealthConnect_DS_EDA.ipynb. Key structural facts:

Attribute	Value
Rows (appointments)	5,000
Columns	18
Unique patients	1,696 (average ~2.95 appointments per patient)
Date range — booking_date	7 Nov 2024 to 27 Jun 2026
Date range — appointment_date	1 Jan 2025 to 30 Jun 2026
Target field	appointment_outcome — Attended (46.3%), No-Show (48.5%), Cancelled (5.3%)

The 18 fields fall into the categories anticipated in the Week 4 brief: identifiers (appointment_id, patient_id), demographics (gender, age, age_group), appointment details (appointment_type, appointment_day, appointment_time), booking information (booking_date, appointment_date, booking_lead_days), patient history (previous_appointments, previous_no_shows), reminders (reminder_sent, reminder_channel), logistics (distance_to_clinic_km, waiting_time_minutes), and the outcome (appointment_outcome).

3.2 Data Quality Assessment — Results

The planned checks (missing values, duplicates, logical consistency, date-logic validation) have now been run against the real file. Results:

Check	Result
Missing values	reminder_channel: 1,366 missing (27.3); distance_to_clinic_km: 90 missing (1.8%); waiting_time_minutes: 60 missing (1.2%). All other fields fully populated.
Duplicate records	0 fully duplicated rows; 0 duplicated appointment_id values.
Data type consistency	Dates parse cleanly in ISO/standard format; numeric fields contain no stray text; categorical fields use a small, consistent set of labels.
Logical consistency	0 rows where previous_no_shows exceeds previous_appointments; 0 rows where booking_date is after appointment_date; booking_lead_days matches the calculated date difference in every row.

Outlier screening	distance_to_clinic_km ranges 0.5–45 km (median 8.7); waiting_time_minutes ranges 2–68 minutes (median 24); no negative or implausible values found.
Class balance	Close to balanced: 48.5% No-Show vs. 46.3% Attended vs. 5.3% Cancelled — materially different from the imbalanced-minority-class assumption made before the data was available.
Leakage risk	waiting_time_minutes is only known once an appointment is underway and is excluded as a model feature for that reason.

Note on reminder_channel missingness: this is not a data quality defect — it is missing precisely and only when reminder_sent = “No” (verified by cross-tabulation), i.e. no reminder means no channel to record. It is a structural relationship between two fields, not a random gap. Filled empty gaps with “None”.

Note on repeated patients: each patient appears in the dataset an average of ~3 times. This matters for modelling: a random row-level train/test split could place the same patient's appointments in both sets, leaking patient-specific behaviour and overstating model performance. A patient-grouped or chronological split is required.

4. Proposed Target Variable

Proposed target: a binary variable, e.g. No-Show , encoded as:

Value	Meaning
1 / “Yes”	If appointment_outcome = ‘No-show’
0 / “No”	If appointment_outcome = ‘Attended’

5. Potential Input Features

EDA tested every candidate field against the no-show outcome. Findings are summarised below, ranked by observed relationship strength rather than assumption.

Category	Candidate Features	Evidence from Week 4 EDA
Scheduling (strongest)	booking_lead_days	Strongest signal found: no-show rate rises from ~28% (0–7 days lead) to ~68% (45–60 days lead) — a clear, monotonic relationship. Correlation with no-show ≈ 0.27.
Patient history (strong)	previous_no_shows (and a derived previous_no_show_rate)	No-show rate rises from ~44% at 0 prior no-shows to ~68% at 3 prior no-shows. Correlation with no-show ≈ 0.13.

Logistics (weak-moderate)	distance_to_clinic_km	Weak positive relationship (correlation ≈ 0.06); retained as a candidate feature pending further testing in the modelling stage.
Demographics (weak)	age, age_group, gender	Weak/flat relationship with outcome (no-show rates for all age bands and genders fall within a narrow $\sim 44\text{--}51\%$ band); retained as low-priority candidates, not primary drivers.
Appointment attributes (weak)	appointment_type, appointment_day, appointment_time	Flat relationship with outcome across categories (within a $\sim 4\text{--}6$ point band); unlikely to be strong standalone predictors but may add value in combination with other features.
Engagement (weak, operationally relevant)	reminder_sent, reminder_channel	Weak relationship with the outcome as recorded; still operationally important to the clinic (see Section 8), and reminder_channel's missingness is structural (only absent when no reminder was sent), not random.
Excluded by design	waiting_time_minutes	Only known once an appointment is already underway — not available at prediction time. Correlation with outcome is effectively zero (≈ 0.00), consistent with it being operationally irrelevant to a pre-appointment prediction. Excluded to avoid leakage.

Overall takeaway: booking_lead_days and prior no-show history are, on current evidence, the two features most likely to drive a useful model; the remaining categorical fields will be retained for the modelling stage but are not expected to be strong predictors individually.

6. Handling of Appointment Outcomes (e.g., Cancellations)

EDA confirmed that appointment_outcome contains exactly three categories — Attended, No-Show, and Cancelled. The proposed handling for modelling:

- **Attended (46.3% of records):** included as the negative class (0).
- **No-Show (48.5% of records):** included as the positive class (1) — this is the primary behaviour the clinic wants to predict and reduce.
- **Cancelled (5.3% of records):** excluded from the primary binary model and analysed as a separate category. This is a small, cleanly separable group and a patient who proactively cancels

is behaviourally different from one who silently fails to attend, so merging it into either class would blur the target definition rather than clarify it.

Because Cancelled is a genuinely small share of the data (5.3%), this decision has limited impact on overall class balance regardless of how it is handled, but excluding it keeps the binary target conceptually clean. This will be confirmed with the Data Analytics track and clinic stakeholders before finalising in the Analysis stage, particularly if the business also wants a combined “no-show or late cancellation” view for operational reporting.

7. Initial Modelling Approach

7.1 Problem Type

Supervised binary classification, using historical labelled appointment outcomes to predict the probability that a future appointment will result in a no-show.

7.1.1. Candidate Model Pipeline

1. **Baseline Model:** Logistic Regression (for interpretability and benchmarking). To establish an interpretable performance baseline and sanity-check feature relationships.
2. **Tree-Based Models:** Random Forest Classifier and XGBoost / LightGBM (to capture non-linear feature interactions).

7.1.2 Metric Selection

- **Primary Metric: ROC-AUC & PR-AUC** (Precision-Recall AUC) due to balanced outcome assessment.
- **Secondary Metric: Recall (Sensitivity)** on the **No-Show** class to minimize False Negatives (unidentified missed appointments).
- **Business Evaluation Metric:** Cost-benefit optimization using custom thresholding to balance reminder cost vs. recovered slot capacity.

7.2 Success Criteria (to be refined with stakeholders)

- The model should meaningfully outperform a naive baseline (e.g., always predicting “attended”, or a simple rule based on historical no-show rate).
- Recall on the no-show class should be prioritised over overall accuracy, given the class imbalance typical of this problem.
- Outputs should be interpretable enough to support a practical intervention (e.g., a risk tier: low/medium/high) rather than a black-box score alone.

8. Key Modelling Considerations, Assumptions, Limitations and Risks

Type	Description
Assumption	The dataset, though fictional, is structured realistically enough to support meaningful feature–target relationships — confirmed by Week 4 EDA, which found a clear, monotonic real-world-consistent pattern between booking lead time and no-shows.
Assumption	The outcome field can be reduced to a usable binary target once cancellations are handled per Section 6; confirmed workable, since Cancelled is a clean, separable ~5.3% of records.
Limitation	Fictional/synthetic data may not fully reflect real-world patient behaviour patterns; this should be caveated in any final presentation of results.
Limitation	EDA to date is univariate/bivariate (one feature vs. outcome at a time); interaction effects between features have not yet been explored and may reveal additional signal.
Revised risk (was assumed, now checked)	Class imbalance is a smaller risk than initially assumed: No-Show (48.5%) and Attended (46.3%) are close to balanced. Standard classification metrics can be used without heavy resampling, though the small Cancelled class (5.3%) still needs careful handling if modelled separately.
Risk	Data leakage: waiting_time_minutes was identified during EDA as available only after an appointment begins and has been explicitly excluded from the candidate feature set for this reason.
Risk	Repeated patients (avg. ~3 appointments each) create a leakage risk specific to model evaluation: a random row-level split could place the same patient in both train and test sets. Mitigation: use a patient-grouped or chronological split.
Risk	Small sample size within certain subgroups (e.g., 5+ prior no-shows, n=3) could limit reliable subgroup analysis or produce unstable estimates in those slices.
Dependency	Alignment with the Data Analytics track on shared business questions and KPI definitions, to avoid duplicated or conflicting analysis.
Ethical / Safety	Demographic features (age, gender, distance) showed weak standalone relationships with the outcome in this dataset, which is reassuring, but fairness/bias checks across subgroups are still planned once a model is trained, since weak marginal relationships do not rule out bias in combination with other features.
Ethical / Safety	Predictions should support staff decision-making (e.g., prioritising reminders) rather than being used to deny or deprioritise care for patients flagged as high-risk.

9. Proposed Focus for Week 5

- Finalise the target variable definition and the treatment of cancellations (Section 6) in discussion with the Data Analytics track.
- Engineer the features prioritised by Week 4 EDA — booking_lead_days, previous_no_shows / previous_no_show_rate — plus the remaining weaker candidates, and encode categoricals.

- Implement a patient-grouped or chronological train/validation/test split to prevent the leakage risk identified in Section 8.
- Produce a first baseline logistic regression model, evaluate it (precision/recall/F1/ROC-AUC), and compare against a naive baseline.
- Extend the notebook (HealthConnect_DS_EDA.ipynb) into the modelling phase and continue documenting in the GitHub repository.